

2024



Improving Location Accuracy for Gas Emergencies Using Amazon Location Service with Cadent

Learn how Cadent in the energy industry simplified locating leaks reported to the National Gas Emergency Service using Amazon Location Service.

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Over 50% reduction

in average call time to report hard-to-



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Improved customer experience

when reporting leaks

Overview

If a person smells a gas leak, they need to be able to report it quickly and accurately. However, if the leak is in an outdoor area without a specific postal address, like a field, passing on the information to emergency responders over the phone can be tricky. If responders aren't sent to the correct location quickly, the public safety risks are increased.

[Cadent](#), a gas distribution network based in Britain that also runs the National Gas Emergency Service line for the United Kingdom, wanted to improve the system for people reporting smells of gas in locations that don't have a specific postal address. By using Amazon Web Services (AWS) and working with [Slalom](#), an [AWS Partner](#), Cadent deployed an application for people who call to report gas emergencies to accurately and quickly pinpoint the location for responders.

Opportunity | Using AWS to Better Locate Outdoor Gas Emergencies for Cadent

In addition to supplying 11 million homes in England with natural gas, Cadent manages the National Gas Emergency Service, a service that members of the public can call 24/7 to report gas emergency situations. The National Gas Emergency Service receives 1.5 million reports per year, and around 900,000 of these incidents require job site visits.

While most incidents take place at a fixed address, about 10 percent of reported gas leaks take place outside in public areas, and about 10 percent of those outdoor locations are difficult for callers to explain. Someone smelling gas in a field or along a public path or road could have trouble describing the location precisely enough for the operators to send work crews to the right place. The previous methods that Cadent used to pinpoint locations relied on how accurately the caller could relay information back to the operator over the phone. "Getting a location wrong could result in serious safety risks for people," says Dan Edwards, head of customer center operations for Cadent. "We wanted to find a better, user-friendly way to locate these emergency situations."

Cadent reached out to several technology service providers, and AWS was the first to respond with a proof of concept in July 2023. Cadent then worked alongside AWS to find a partner to develop the solution, and in October, Cadent began a 4-week engagement with Slalom. The result—the LocateMe application—was delivered to Cadent in November 2023 and, after operator training, went live in early 2024.

Solution | Reducing Call Times by 50% Using Amazon Location Service

Now, when a person calls the National Gas Emergency Service and reports a smell of gas in a location without a postal address, the operator asks the person permission to send a text message to define the location of the emergency. If the caller agrees, they receive an SMS with a link to open the LocateMe app, which asks for their GPS location. The app then shows the caller's current location on a map, and the caller can confirm if the problem is at that location or edit the location to pinpoint where the gas smell is. The app sends the location back to the operator, who can copy it directly into the system. This supplies the emergency team with precise GPS coordinates.

Using the LocateMe app, average call times to report gas emergencies at locations without postal addresses have been reduced from 12 minutes to 5–6 minutes. With shorter calls, operators can alert response teams to deal with each emergency more quickly. “Ultimately, whenever there’s a gas leak, our job is to get engineers to that specific site to investigate it, resolve it, and make the situation safe,” says Edwards. “Now, we can get engineers to the right location 5 or 6 minutes faster.”

To display map locations in LocateMe, the app uses [Amazon Location Service](#), a location-based service that developers can use to add geospatial data and location functionality to applications. (See figure 1.)

“The map in LocateMe, powered by Amazon Location Service, is intuitive,” says Edwards. “For the end user, it gives confidence that the correct location has been passed on to our agents.”

Slalom built the application for Cadent using coding languages and tooling that Cadent was already familiar with. Thus, Cadent, which now owns and runs the application, can maintain and further develop it without making major changes to its systems or team. To build the application quickly using Cadent’s preferred technology, Slalom used [AWS App Runner](#)—a fully managed application service used to build, deploy, and run web applications and API services without prior infrastructure or container experience. “Slalom brought not only quick build time and quality from its architects and engineers but also worked with us as a stakeholder to really understand what the intention was for this app,” says Edwards. “Slalom actually delivered a far better product than what we’d asked for because its team really understood what we were aiming to achieve.”

The solution has been reviewed by various key stakeholders and regulatory authorities in the United Kingdom, including the Health and Safety Executive. “Every person that I’ve demonstrated this solution to is amazed by how much better it is than what we used before,” says Edwards. “The success of the solution has made it simple to use, demonstrating our commitment to continuous improvement and enhancing customer safety.”

Architecture Diagram





AWS Services Used

Amazon Location Service

Amazon Location Service makes it easy for developers to add location functionality, such as maps, points of interest, geocoding, routing, tracking, and geofencing, to their applications without sacrificing data security and user privacy.

[Learn more »](#)

AWS App Runner

AWS App Runner builds and deploys web applications automatically, load balances traffic with encryption, scales to meet your traffic needs, and allows for the configuration of how services are accessed and communicate with other AWS applications in a private Amazon VPC.

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